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Assignment 11

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Simple Linear Regression: Information, Scores, and Geometry

Assignment 11 takes the simple linear regression model worked at the start of class and reads it three ways at once—through the information matrix, through the score vector, and through the orthogonal geometry of n-dimensional space—showing that the three tell one story. It is the natural sequel to Assignment 10: there the design had two group-indicator columns; here it has an intercept and a genuine covariate.

The model. The data are $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where the $n \times 2$ design matrix carries a column of ones and a column of the covariate,

$$\mathbf{X} = \begin{pmatrix} \mathbf{1} & \mathbf{x} \end{pmatrix}, \qquad \mathbf{1} = (1, \ldots, 1)^\top, \qquad \mathbf{x} = (x_1, \ldots, x_n)^\top, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix},$$

with β_0 the coefficient on the ones and β_1 the coefficient on the x's. The errors are normal with mean zero and a *known* common variance, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, so σ^2 is not a parameter to be estimated. Throughout, $\bar{x} = \frac{1}{n}\sum_i x_i$, the centered covariate is $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}$, and

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}} = \sum_i x_i^2 - n\bar{x}^2.$$

Two facts recur: $\mathbf{X}^\top \mathbf{X} = \left(\begin{smallmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{smallmatrix}\right)$ has determinant $n\sum_i x_i^2 - (n\bar{x})^2 = nS_{xx}$, and $\mathbf{1}^\top \tilde{\mathbf{x}} =$

$\sum_i (x_i - \bar{x}) = 0$, so the ones and the centered covariate are orthogonal.

Part 1

Compute the variance–covariance matrix of the estimated regression coefficients and compute the information matrix. Compare them and say what you find.

SOLUTION:

The covariance matrix. The least squares (and, under normality, maximum likelihood) estimator is $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$, linear in $\mathbf{Y}$, so with $\mathrm{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}_n$,

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\sigma^2\mathbf{I}_n)\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{\sigma^2}{nS_{xx}}\begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix}.$$

Reading off the entries (and using $\sum_i x_i^2 = S_{xx} + n\bar{x}^2$),

$$\mathrm{Var}(\hat{\beta}_0) = \sigma^2\left(\tfrac{1}{n} + \tfrac{\bar{x}^2}{S_{xx}}\right), \qquad \mathrm{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \qquad \mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2\bar{x}}{S_{xx}}.$$

The information matrix. With σ^2 known, the log likelihood is $\ell(\boldsymbol{\beta}) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$, whose gradient and Hessian in $\boldsymbol{\beta}$ are

$$\mathbf{l}(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \frac{1}{\sigma^2}\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}), \qquad \frac{\partial^2 \ell}{\partial \boldsymbol{\beta}\,\partial \boldsymbol{\beta}^\top} = -\frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}.$$

The Hessian is constant, so the Fisher information—minus its expectation—is

$$\mathcal{I}(\boldsymbol{\beta}) = -\,\mathrm{E}\left[\frac{\partial^2 \ell}{\partial \boldsymbol{\beta}\,\partial \boldsymbol{\beta}^\top}\right] = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X} = \frac{1}{\sigma^2}\begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}.$$

(The same matrix arises from the outer-product form: $\mathrm{E}\{\mathbf{l}\mathbf{l}^\top\} = \sigma^{-4}\mathbf{X}^\top \mathrm{E}\{\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top\}\mathbf{X} = \sigma^{-4}\mathbf{X}^\top(\sigma^2\mathbf{I}_n)\mathbf{X} = \sigma^{-2}\mathbf{X}^\top\mathbf{X}$.)

What I find. The two matrices are exact inverses:

$$\boxed{\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \left(\tfrac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}\right)^{-1} = \mathcal{I}(\boldsymbol{\beta})^{-1}.}$$

The covariance of the coefficient estimates *is* the inverse information. This is the multivariate Cramér–Rao bound attained with equality, and—because the Hessian here is nonrandom—attained not merely asymptotically but *exactly*, in finite samples. It is the matrix version of the scalar efficiency the course met in Module II (Assignments 5 and 6), where in the normal model the maximum likelihood estimator sat right on the information bound. The reason is structural: the normal linear model is a full-rank exponential family in which $\hat{\boldsymbol{\beta}}$ is a linear function of the sufficient statistic $\mathbf{X}^\top\mathbf{Y}$, and for such estimators the bound is an equality. Part 2 proves the equality directly from the scores, without ever inverting a matrix by hand. ■

Part 2

Show that the least squares estimates minus their expectations can be written as $\mathbf{M}\mathbf{l}$, where $\mathbf{M}$ is the inverse of the information matrix and $\mathbf{l}$ is the vector of scores. Use this to conclude that the variance–covariance matrix of the coefficient estimates is the inverse information.

SOLUTION:

The representation. Evaluate the score at the true $\boldsymbol{\beta}$; there $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\varepsilon}$, so

$$\mathbf{l} = \mathbf{l}(\boldsymbol{\beta}) = \frac{1}{\sigma^2}\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = \frac{1}{\sigma^2}\mathbf{X}^\top\boldsymbol{\varepsilon}.$$

The estimator is unbiased, $\mathrm{E}\{\hat{\boldsymbol{\beta}}\} = \boldsymbol{\beta}$, and its deviation from the truth is

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} - \boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}.$$

Now substitute $\mathbf{X}^\top\boldsymbol{\varepsilon} = \sigma^2\mathbf{l}$ from the score above:

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\big(\sigma^2\mathbf{l}\big) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{l} = \mathcal{I}(\boldsymbol{\beta})^{-1}\mathbf{l},$$

since $\mathcal{I}^{-1} = \big(\sigma^{-2}\mathbf{X}^\top\mathbf{X}\big)^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$. Writing $\mathbf{M} = \mathcal{I}^{-1}$,

$$\boxed{\hat{\boldsymbol{\beta}} - \mathrm{E}\{\hat{\boldsymbol{\beta}}\} = \mathbf{M}\,\mathbf{l}, \qquad \mathbf{M} = \mathcal{I}(\boldsymbol{\beta})^{-1}.}$$

The coefficient error is a fixed linear image of the score—the inverse information steering the score into the parameter's coordinates.

The covariance, again. The score has mean zero and covariance equal to the information, $\mathrm{Cov}(\mathbf{l}) = \mathcal{I}$ (that is the identity $\mathrm{Var}(\mathbf{l}) = \mathrm{E}\{\mathbf{l}\mathbf{l}^\top\} = \mathcal{I}$ used in Part 1). Since

$\mathbf{M} = \mathcal{I}^{-1}$ is symmetric,

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \mathrm{Cov}(\mathbf{M}\mathbf{l}) = \mathbf{M}\ \mathrm{Cov}(\mathbf{l})\,\mathbf{M}^\top = \mathcal{I}^{-1}\,\mathcal{I}\,\mathcal{I}^{-1} = \mathcal{I}^{-1}.$$

So $\boxed{\mathrm{Cov}(\hat{\boldsymbol{\beta}}) = \mathcal{I}(\boldsymbol{\beta})^{-1}}$ falls out of the representation in one line, with no matrix inverted by hand—the same conclusion as Part 1, now seen as a consequence of the score's own covariance being the information. The middle $\mathcal{I}$ cancels one of the two $\mathcal{I}^{-1}$ factors; this cancellation is exactly why an efficient estimator's covariance collapses to the inverse information rather than the larger sandwich a general linear estimator would carry. ■

Part 3

What linear combination of scores gives the estimate of the coefficient on the x's? What is its covariance with the score for the coefficient on the ones, and with the score for the coefficient on the x's? What do you make of that?

Solution:

Write the two scores by their columns: $\mathbf{l} = (l_0, l_1)^\top$ with

$$l_0 = \frac{1}{\sigma^2}\mathbf{1}^\top\boldsymbol{\varepsilon} = \frac{1}{\sigma^2}\sum_i \varepsilon_i \quad \text{(ones)}, \qquad l_1 = \frac{1}{\sigma^2}\mathbf{x}^\top\boldsymbol{\varepsilon} = \frac{1}{\sigma^2}\sum_i x_i\varepsilon_i \quad (x\text{'s}).$$

The combination. By Part 2, $\hat\beta_1 - \beta_1 = (\mathbf{M}\mathbf{l})_2$, the second row of $\mathbf{M} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{\sigma^2}{nS_{xx}}\begin{pmatrix}\sum_i x_i^2 & -n\bar x \\ -n\bar x & n\end{pmatrix}$ applied to $\mathbf{l}$. That second row is $\frac{\sigma^2}{nS_{xx}}(-n\bar x,\ n) = (-\frac{\sigma^2\bar x}{S_{xx}},\ \frac{\sigma^2}{S_{xx}})$, so

$$\boxed{\hat\beta_1 - \beta_1 = \frac{\sigma^2}{S_{xx}}\big(l_1 - \bar x\, l_0\big).}$$

The slope is recovered from the scores by removing $\bar x$ times the intercept score and scaling by σ^2/S_{xx}. (In errors this reads $\frac{\sigma^2}{S_{xx}} \cdot \frac{1}{\sigma^2}(\mathbf{x} - \bar x\mathbf{1})^\top\boldsymbol{\varepsilon} = \tilde{\mathbf{x}}^\top\boldsymbol{\varepsilon}/S_{xx}$, the form Part 4 needs.)

Its covariances. The scores' covariances are the entries of $\mathcal{I}$: $\mathrm{Var}(l_0) = n/\sigma^2$, $\mathrm{Cov}(l_0, l_1) = n\bar x/\sigma^2$, $\mathrm{Var}(l_1) = \sum_i x_i^2/\sigma^2$. Hence

$$\mathrm{Cov}\left(\hat\beta_1 - \beta_1,\ l_0\right) = \frac{\sigma^2}{S_{xx}}\Big[\mathrm{Cov}(l_1, l_0) - \bar x\,\mathrm{Var}(l_0)\Big] = \frac{\sigma^2}{S_{xx}}\left[\frac{n\bar x}{\sigma^2} - \bar x \cdot \frac{n}{\sigma^2}\right] = \boxed{0},$$

$$\mathrm{Cov}\left(\hat\beta_1 - \beta_1,\ l_1\right) = \frac{\sigma^2}{S_{xx}}\Big[\mathrm{Var}(l_1) - \bar x\,\mathrm{Cov}(l_0, l_1)\Big] = \frac{\sigma^2}{S_{xx}} \cdot \frac{\sum_i x_i^2 - n\bar x^2}{\sigma^2} = \frac{\sigma^2}{S_{xx}} \cdot \frac{S_{xx}}{\sigma^2} = \boxed{1}.$$

These are no accident. In general $\mathrm{Cov}(\hat{\boldsymbol\beta} - \boldsymbol\beta, \mathbf{l}) = \mathrm{Cov}(\mathbf{M}\mathbf{l}, \mathbf{l}) = \mathbf{M}\,\mathcal{I} = \mathcal{I}^{-1}\mathcal{I} = \mathbf{I}_2$, the identity: each coefficient's error has covariance 1 with its own score and 0 with the other.

What I make of it. The unit covariance with l_1 is a normalization—the combination is calibrated to move one-for-one with its own score. The *zero* covariance with l_0 is the content: the slope estimate's error is uncorrelated with the intercept's score, and under normality uncorrelated means independent. Estimating β_1 has been *orthogonalized* against the intercept—the slope draws only on the part of the data the ones cannot explain. This is the algebraic shadow of the geometry in Parts 4–5, where $\hat{\beta}_1 - \beta_1$ turns out to live along $\tilde{\mathbf{x}}$ (the covariate with its mean removed), a direction orthogonal to $\mathbf{1}$, while the intercept score $l_0 \propto \mathbf{1}^\top \boldsymbol{\varepsilon}$ lives along $\mathbf{1}$; orthogonal directions of a Gaussian vector are uncorrelated, hence independent. ■

Part 4

Can you write the least squares estimate of the coefficient on the x's as the coefficient of the projection of $\mathbf{Y}$ onto the span of a vector in n-dimensional space? Show that vector is a constant times the residual from the projection of $\mathbf{x}$ onto the span of $\mathbf{1}$. What is the constant?

SOLUTION:

The residual of $\mathbf{x}$ on the ones. Projecting the covariate onto span$(\mathbf{1})$ uses $\mathsf{P}_{\mathbf{1}} = \mathbf{1}(\mathbf{1}^\top\mathbf{1})^{-1}\mathbf{1}^\top = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, giving fitted part $\mathsf{P}_{\mathbf{1}}\mathbf{x} = \bar{x}\mathbf{1}$ and residual

$$\tilde{\mathbf{x}} = (\mathbf{I}_n - \mathsf{P}_{\mathbf{1}})\mathbf{x} = \mathbf{x} - \bar{x}\mathbf{1}, \qquad \tilde{\mathbf{x}}_i = x_i - \bar{x}.$$

This centered covariate is the "portion of $\mathbf{x}$ orthogonal to $\mathbf{1}$": $\mathbf{1}^\top\tilde{\mathbf{x}} = 0$ and $\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}} = S_{xx}$.

The slope is a projection coefficient. Because $\sum_i(x_i - \bar{x}) = 0$, the numerator of the usual slope formula may be written against $\mathbf{Y}$ directly,

$$\hat{\beta}_1 = \frac{\sum_i(x_i-\bar{x})(Y_i-\bar{Y})}{\sum_i(x_i-\bar{x})^2} = \frac{\sum_i(x_i-\bar{x})Y_i}{\sum_i(x_i-\bar{x})^2} = \frac{\tilde{\mathbf{x}}^\top\mathbf{Y}}{\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}}.$$

But $\tilde{\mathbf{x}}^\top\mathbf{Y}/\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}$ is exactly the coefficient in the orthogonal projection of $\mathbf{Y}$ onto the line span$(\tilde{\mathbf{x}})$:

$$\mathsf{P}_{\tilde{\mathbf{x}}}\mathbf{Y} = \frac{\tilde{\mathbf{x}}^\top\mathbf{Y}}{\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}}\,\tilde{\mathbf{x}} = \hat{\beta}_1\,\tilde{\mathbf{x}}.$$

So

$$\boxed{\hat{\beta}_1 = \frac{\tilde{\mathbf{x}}^\top\mathbf{Y}}{\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}} \ \text{ is the coefficient of } \ \mathsf{P}_{\mathrm{span}(\tilde{\mathbf{x}})}\mathbf{Y} = \hat{\beta}_1\,\tilde{\mathbf{x}}, \quad \tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1} = (\mathbf{I}_n - \mathsf{P}_{\mathbf{1}})\mathbf{x}.}$$

The vector spanning that line is precisely the residual of $\mathbf{x}$ on the ones—the constant of proportionality is

$$\boxed{1}$$

No rescaling is needed: the projection coefficient onto $\operatorname{span}(c\,\tilde{\mathbf{x}})$ equals $\tilde{\mathbf{x}}^\top \mathbf{Y}/(c\,\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}) = \hat{\beta}_1/c$, which returns $\hat{\beta}_1$ only at $c = 1$. The centered covariate is thus the unique spanning vector whose projection coefficient is the slope itself. (Equivalently, if one prefers $\hat{\beta}_1$ as a plain linear functional $\hat{\beta}_1 = \mathbf{a}^\top\mathbf{Y}$, the weight vector is $\mathbf{a} = \tilde{\mathbf{x}}/S_{xx}$—the same residual, now scaled by $1/S_{xx}$. The two readings differ only in whether the length S_{xx} is carried by the vector or by the coefficient.) That the numerator could drop $\bar{Y}$ is the whole trick: regressing on a centered covariate needs no separate centering of $\mathbf{Y}$, because $\tilde{\mathbf{x}}$ is already orthogonal to the ones that carry $\bar{Y}$. ■

Part 5

We are viewing n-dimensional space as the span of three subspaces: one spanned by $\mathbf{1}$, one by the portion of $\mathbf{x}$ orthogonal to $\mathbf{1}$, and the rest the orthogonal complement of those two. Find the relationship between the least squares estimator of the coefficient on the x's and the projection onto the second subspace. In terms of the columns of ones, of x's, and of errors, what is that projection?

SOLUTION:

The decomposition. The three subspaces

$$V_1 = \operatorname{span}(\mathbf{1}), \qquad V_2 = \operatorname{span}(\tilde{\mathbf{x}}), \qquad V_3 = \big(V_1 \oplus V_2\big)^{\perp} = \big(\text{column space of } \mathbf{X}\big)^{\perp}$$

are mutually orthogonal and decompose the whole space, $\mathbb{R}^n = V_1 \oplus V_2 \oplus V_3$, with dimensions 1, 1, and $n-2$. Orthogonality is by construction: $\mathbf{1}^\top \tilde{\mathbf{x}} = 0$, and V_3 is defined as the complement. Because $\operatorname{span}(\mathbf{1}, \mathbf{x}) = \operatorname{span}(\mathbf{1}, \tilde{\mathbf{x}})$, the first two subspaces together are exactly the model's column space; V_3 is the error space left over.

The slope is the V_2-coordinate of $\mathbf{Y}$. The projection of $\mathbf{Y}$ onto the second subspace is, by Part 4,

$$\mathsf{P}_{V_2}\mathbf{Y} = \frac{\tilde{\mathbf{x}}^\top \mathbf{Y}}{\tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}}\, \tilde{\mathbf{x}} = \hat{\beta}_1\, \tilde{\mathbf{x}}.$$

So the relationship is exact and clean:

$$\boxed{\mathsf{P}_{V_2}\mathbf{Y} = \hat{\beta}_1\, \tilde{\mathbf{x}}} \qquad \Longleftrightarrow \qquad \hat{\beta}_1 = \text{the coordinate of } \mathsf{P}_{V_2}\mathbf{Y} \text{ along } \tilde{\mathbf{x}}.$$

The least squares slope is nothing but the coordinate of $\mathbf{Y}$'s V_2-component along the basis vector $\tilde{\mathbf{x}}$. The intercept subspace V_1 and the residual subspace V_3 contribute nothing to it.

The projection in terms of ones, x's, and errors. Split the covariate as $\mathbf{x} = \bar{x}\mathbf{1} + \tilde{\mathbf{x}}$ and write the mean structure with it:

$$\mathbf{Y} = \beta_0\mathbf{1} + \beta_1\mathbf{x} + \boldsymbol{\varepsilon} = (\beta_0 + \beta_1\bar{x})\mathbf{1} + \beta_1\tilde{\mathbf{x}} + \boldsymbol{\varepsilon}.$$

The three terms sit in V_1, V_2, and (after $\boldsymbol{\varepsilon}$ is itself split) all three subspaces. Applying P_{V_2} and using $\mathsf{P}_{V_2}\mathbf{1} = \mathbf{0}$ (as $\mathbf{1} \perp \tilde{\mathbf{x}}$) and $\mathsf{P}_{V_2}\tilde{\mathbf{x}} = \tilde{\mathbf{x}}$,

$$\mathsf{P}_{V_2}\mathbf{Y} = \underbrace{\beta_1\,\tilde{\mathbf{x}}}_{\text{signal, from the } x \text{ column}} + \underbrace{\frac{\tilde{\mathbf{x}}^\top\boldsymbol{\varepsilon}}{\tilde{\mathbf{x}}^\top\tilde{\mathbf{x}}}\,\tilde{\mathbf{x}}}_{\text{noise, from the error column}} = \Big(\beta_1 + \frac{\tilde{\mathbf{x}}^\top\boldsymbol{\varepsilon}}{S_{xx}}\Big)\tilde{\mathbf{x}}.$$

Matching this with $\mathsf{P}_{V_2}\mathbf{Y} = \hat{\beta}_1\tilde{\mathbf{x}}$ reads off the estimator and its error at once:

$$\boxed{\hat{\beta}_1 = \beta_1 + \frac{\tilde{\mathbf{x}}^\top\boldsymbol{\varepsilon}}{S_{xx}}, \qquad \hat{\beta}_1 - \beta_1 = \frac{\tilde{\mathbf{x}}^\top\boldsymbol{\varepsilon}}{S_{xx}} = \frac{(\mathbf{x} - \bar{x}\mathbf{1})^\top\boldsymbol{\varepsilon}}{S_{xx}}.}$$

In words: the projection of $\mathbf{Y}$ onto V_2 is β_1 times the centered covariate $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}$ (the true signal) plus the projection of the error column $\boldsymbol{\varepsilon}$ onto that same direction (the noise). The signal fixes the target $\beta_1\tilde{\mathbf{x}}$; the noise supplies the entire sampling error $\hat{\beta}_1 - \beta_1$, and it enters only through the component of $\boldsymbol{\varepsilon}$ lying in V_2.

This closes the loop with Part 3. The slope's error $\hat{\beta}_1 - \beta_1 = \tilde{\mathbf{x}}^\top\boldsymbol{\varepsilon}/S_{xx}$ lives along V_2, whereas the intercept score $l_0 = \mathbf{1}^\top\boldsymbol{\varepsilon}/\sigma^2$ lives along V_1; $V_1 \perp V_2$ forces $\mathrm{Cov}(\hat{\beta}_1 - \beta_1, l_0) = 0$—precisely the zero found algebraically in Part 3, now visibly a statement about orthogonal directions in $\mathbb{R}^n$. The known-variance regression has resolved into three orthogonal pieces: the ones fix the level, the centered covariate carries the slope, and the remaining $n-2$ dimensions hold the error that—were σ^2

unknown—Assignment 10 would have used to estimate it. ■

References

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